Weekly low-stakes quizzing and final-exam performance in introductory biology: a cluster-randomised trial

Chidi Okafor, Lena Marsh, Tomas Ruiz

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Abstract

We randomised 14 sections of an introductory biology course to weekly low-stakes quizzes or to the usual homework schedule. The trial enrolled 2,186 students. Students in quizzing sections scored 4.1 percentage points higher on the common final exam.

1. Methods

Design: cluster-randomised controlled trial, with course section as the unit of randomisation. Participants: 2,186 students enrolled in 14 sections (7 per arm) in autumn 2021 and spring 2022. Intervention: a ten-item online quiz each week counting for 5% of the grade; control sections kept the usual weekly homework. Outcome: score on a common final exam marked blind to section.

2. Results

Final-exam scores were 4.1 percentage points higher in quizzing sections (mixed model accounting for clustering; 95% CI 1.9 to 6.3; ICC 0.04).

3. Limitations

The trial ran at a single university. Instructors knew their section's allocation, and quiz completion was not recorded for 9% of students.

Fictional paper generated to test evidence-table extraction. Authors, journal, data and results are invented.